AN EXPLORATION OF CHAOS

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A Special Recursive Sequence

A few months ago, my math teacher, Mr. Shah, told me about this simple looking recursive sequence:

$$x_{n+1} = rx_n(1 - x_n)$$

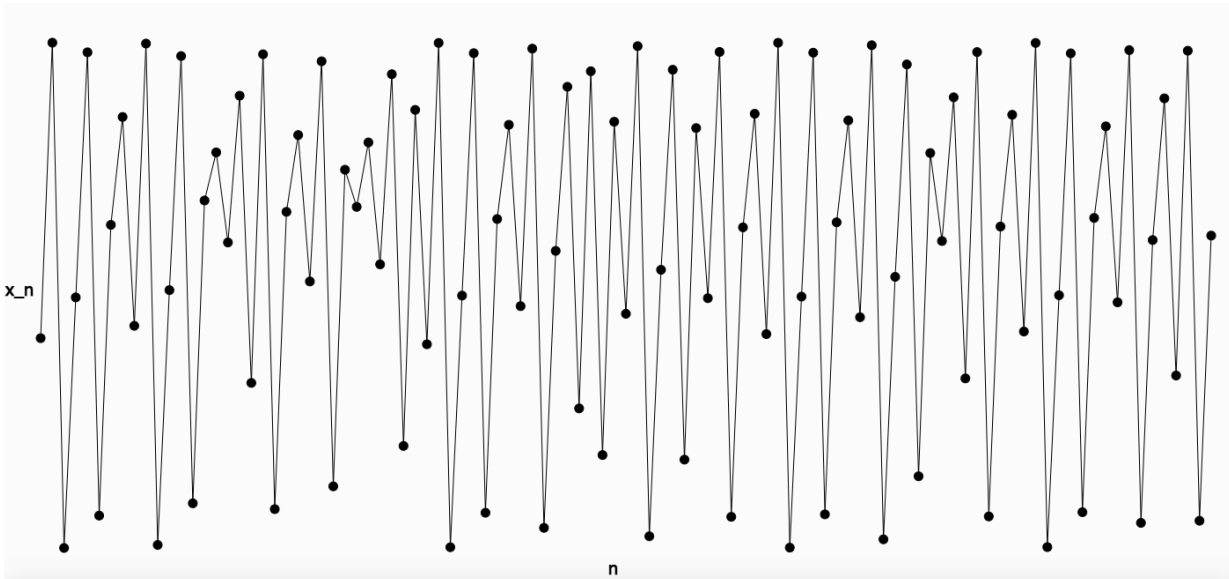
I decided to do some investigating. My first instinct was to write a p5.js program to qualitatively visualize how this sequence evolves, depending on which values I pick for r and the sequence's first term (x_1). As per Mr. Shah's advice, I only looked at r values between 0 and 4, and first terms between 0 and 1. My program graphed the first 100 terms of the sequence.

The results were interesting.

r = 0.5 and x_1 = 0.5

This was what I had expected. However, something very strange happened when I increased the value of r.

<u>r = 3.8 and x_1 = 0.5:</u>



When I first got a glimpse of this utter mayhem, I immediately thought that maybe something had gone wrong with my program… but I realized that my program was functioning just fine. It was performing the same exact operations as it had for the smaller r value, but for some reason, there wasn't any discernible pattern.

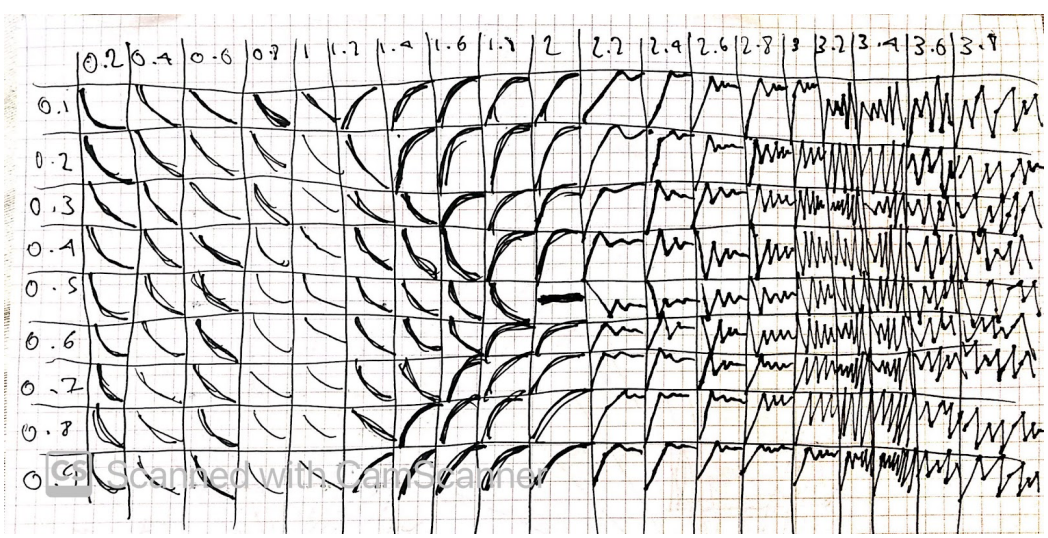
At this point, my biggest question was this:

How can such a seemingly <u>simple</u> recursive sequence behave in such a <u>chaotic</u> way?

In order to get some insight into how changing the initial conditions of the equation might affect the sequence's behavior, I created a chart with sketches of the sequence for different values of r and x_1.

r

x_1



Scanned with CamScanner

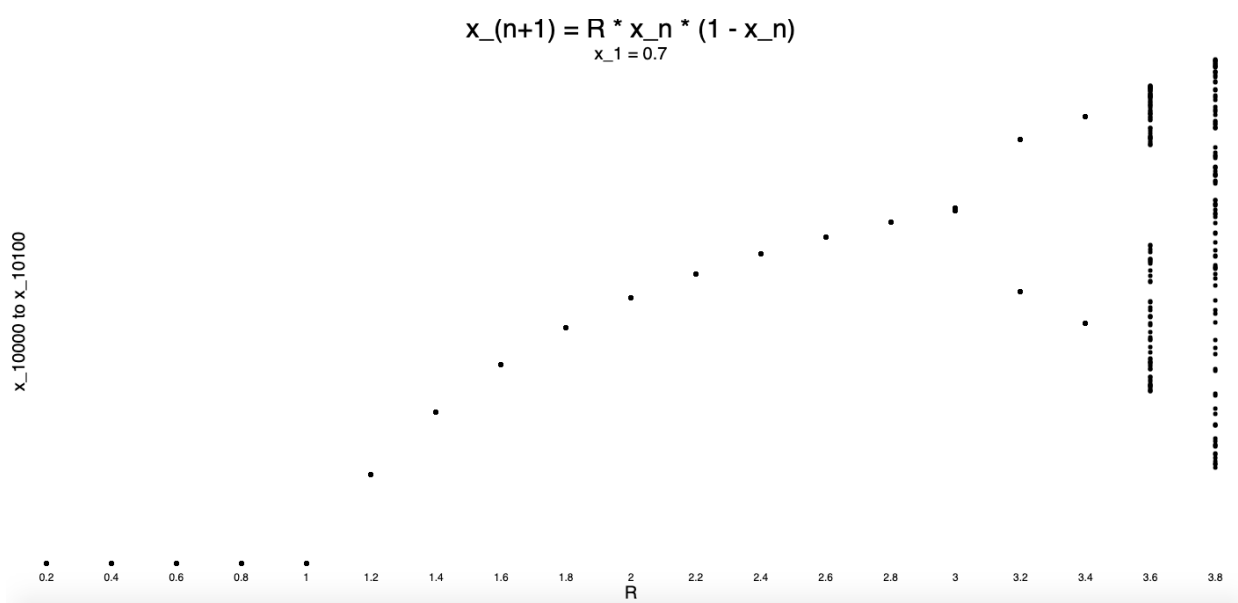
There were two important things I took away from this chart:

1. The first term of the sequence (x_1) <u>does not</u> seem to impact the behavior of the sequence.
2. The value of *r,* however, <u>does</u> impact the behavior.
    a. When *r* is small, the behavior of the graph is fairly simple. However, as *r* increases*,* we start to see the sequence bounce back and forth. Starting around r=3.6, it seems totally random.

In order to find out a) exactly what is happening as *r* increases, and b) for precisely what values of *r* this strange behavior emerges, I needed a new way to visualize the sequence.

<u>A Different Type of Graph</u>

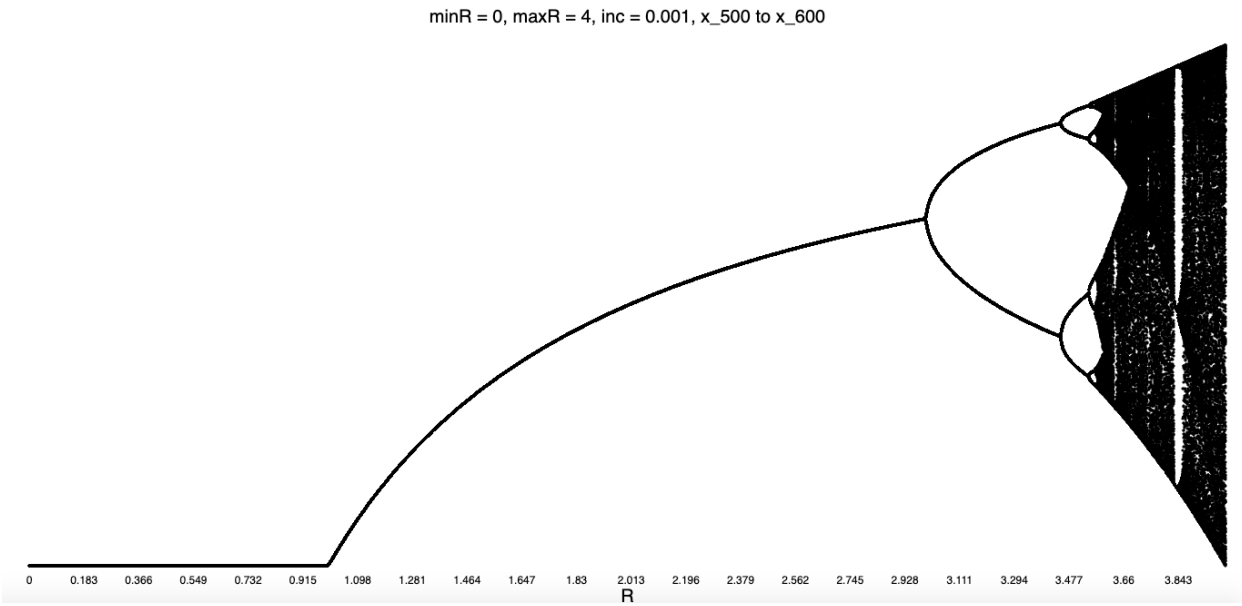
To get a better sense of how the sequence behaves in the <u>long term</u> for different *r* values, I wrote a <u>new program</u> to generate a different type of graph. Instead of visualizing just one sequence with one specific *r* value, this new graph would plot different *r* values (ranging from 0.2 to 3.8) on the x-axis. Also, instead of plotting the first 100 terms of the sequence, this new graph would plot the 10,000th term to the 10,100th term of the sequence on the y-axis for each *r* value. By plotting a bunch of terms that are deep into the sequence, I would get some insight into <u>how</u> the sequence behaves in the long term (i.e. Does it settle on a certain number? Does it bounce between two numbers? Does it do something else?) depending on *r*.



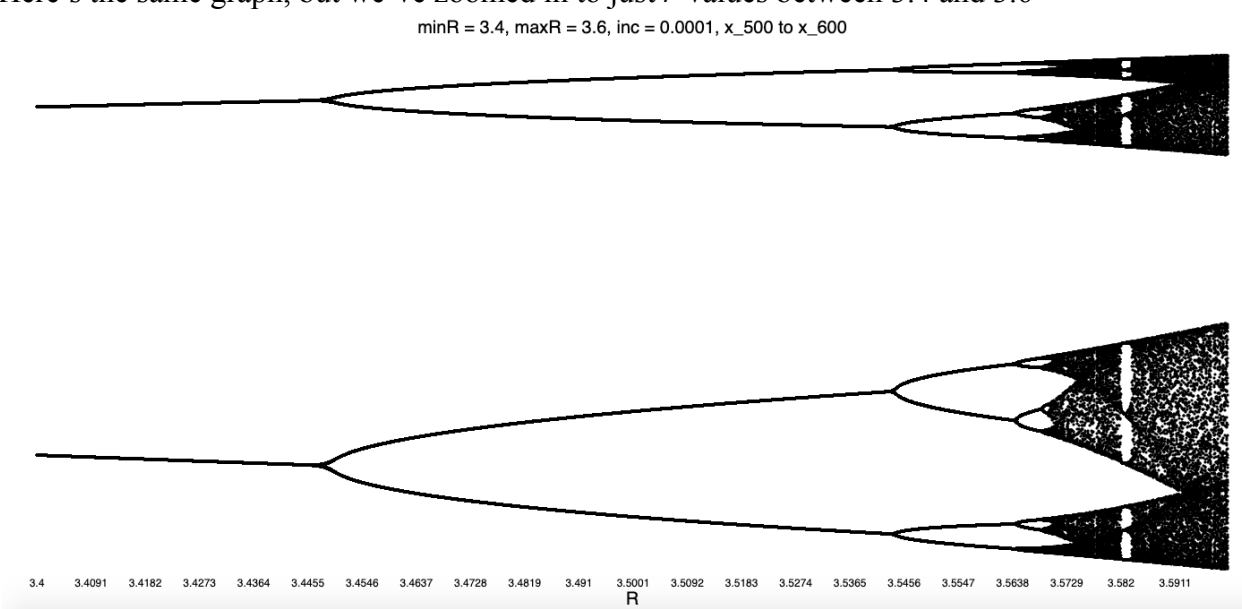
Already, I could see the parallels between this and my previous hand-drawn graph: the sequence settles to one value for low *r*-values, but at around *r* = 3, the long term behavior begins to branch

into two (meaning that the sequence bounces between two different values, rather than settling to one value). And for $r = 3.6$ and $r = 3.8$, the sequence does not settle down to any pattern.

In order to see how exactly this graph goes from something simple (bouncing between two values) to something complicated, I created a more refined graph which plotted the r-values in increments of 0.0001.



Here's the same graph, but we've zoomed in to just r-values between 3.4 and 3.6



With this graph (which is known as the logistic map), we can see how we go from simply bouncing between two numbers ($r = 3.4$) to chaos ($r = 3.6$). The sequence goes from bouncing

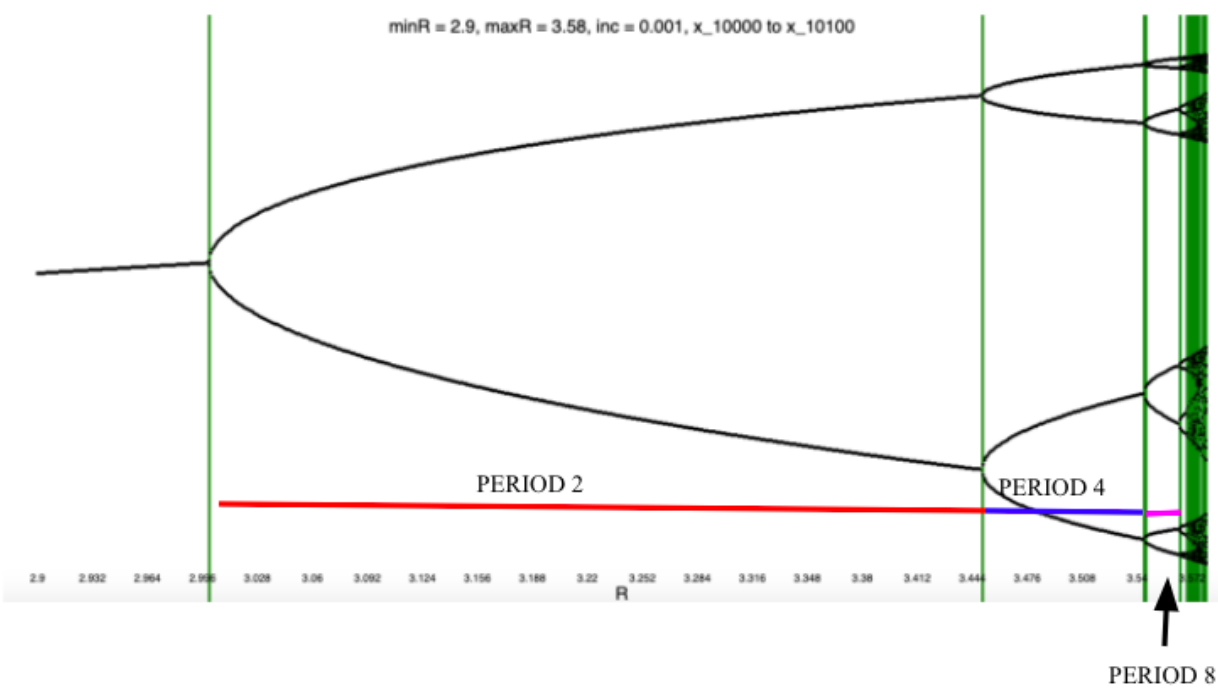
between 2 numbers, to 4 numbers, to 8, to 16, and so on. These “forks" in the graph are known as bifurcations. When the long term sequence bounces between 2 numbers, this is known as a “period 2 cycle,” and when it’s bouncing between 4 numbers, it’s a “period 4 cycle,” and so on. Moreover, the *r*-values at which the graph bifurcates seem closer with each subsequent bifurcation -- the first bifurcation is at around $r = 3.45$, the next one is near $r = 3.54$, then $r = 3.56$, and so on. I later learned that since these bifurcation points gradually get closer and closer together, starting at approximately $r = 3.57$, the logistic map “stops” bifurcating (although it actually bifurcates forever…). The area of the graph after this *r*-value is known as the chaotic region.

<u>Feigenbaum’s Constant</u>

In 1975, physicist Mitchell Feigenbaum made an important discovery about these *r*-values. He noticed that they occur in a certain ratio.

Visualization of the *r*-values at which the logistic map bifurcates

- The vertical green lines highlight where the graph bifurcates
- The horizontal lines represent the different regions of the graph (period 2, period 4, etc.)



Feigenbaum noted that the ratio between the length of the period 2 region and the length of the period 4 region was similar to that of the length of the period 4 region and the length of the period 8 region, and so on. As this process continues, these ratios approach approximately 4.669. This number is known as the Feigenbaum Constant -- the rate at which the logistic map bifurcates.

If we perform the calculations to find the ratios between the first few regions of the logistic map, we can derive Feigenbaum's Constant:

In order to find the length of each region (period 2, period 4, etc.), we need to use the *r*-values where the logistic map bifurcates.

Here are the first few *r*-values where the graph bifurcates:

```
3
3.4495
3.5441
3.5645
```

$$\frac{period\ 2\ length}{period\ 4\ length} = \frac{3.4495-3}{3.5441-3.4495} = 4.7515856237$$

$$\frac{period\ 4\ length}{period\ 8\ length} = \frac{3.5441-3.4495}{3.5645-3.5441} = 4.637254902$$

Actual constant = 4.669201609…

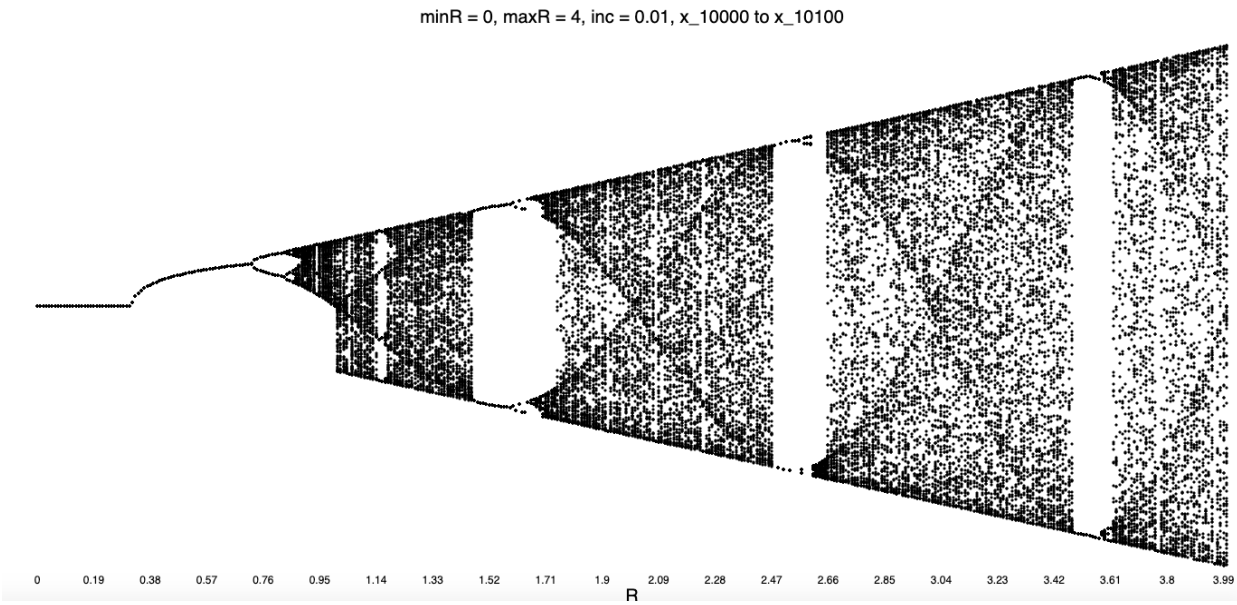
Feigenbaum's Constant is a <u>universal</u> constant, meaning it holds true for not only this specific recursive sequence, but many others.

For example, consider this recursive sequence:

$$x_{n+1} = r * sin(\pi x_n)$$

If we perform the same calculations with its bifurcation points, we also get a ratio near Feigenbaum's Constant.

Bifurcation diagram of a different recursive sequence: $x_{n+1} = r * sin(\pi x_n)$



Note how even in the midst of all the chaotic behavior we see regions of stability!

<u>*R*-values of first few bifurcation points</u>

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0.72
0.8333
0.8587
0.8641
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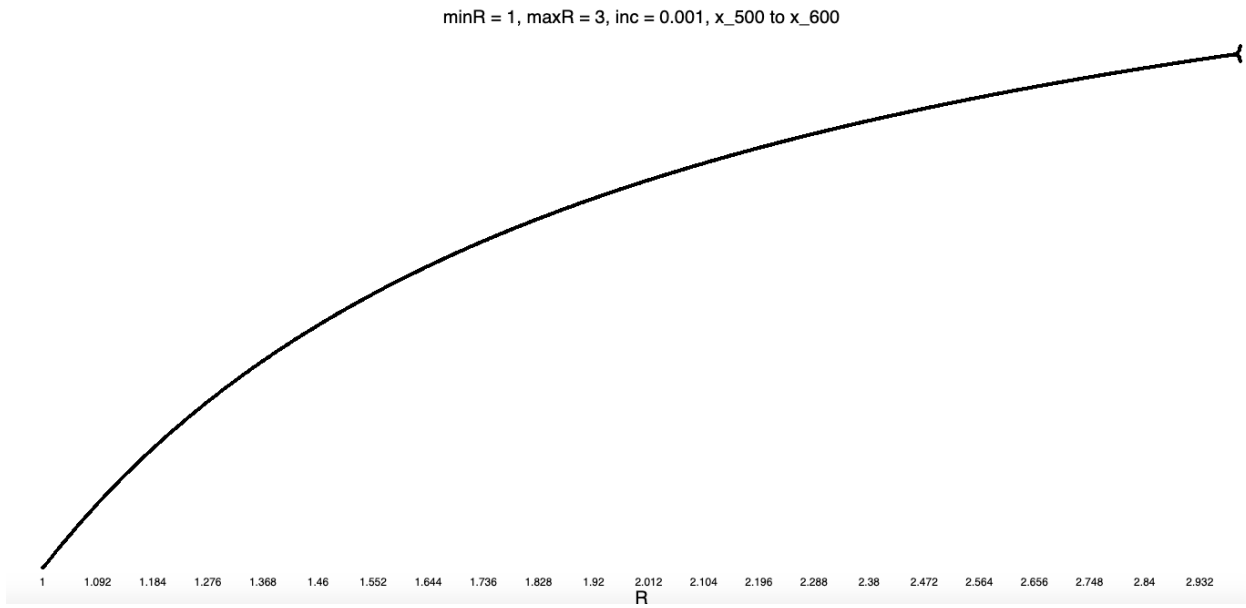
$$\frac{period\ 2\ length}{period\ 4\ length} = \frac{0.8333-0.72}{0.8587-0.8333} = 4.4606299213$$

$$\frac{period\ 4\ length}{period\ 8\ length} = \frac{0.8587-0.8333}{0.8641-0.8587} = 4.7037037037$$

Algebraically Understanding The Logistic Map

Another interesting piece about the logistic map is that many of its features can be explained algebraically.

First consider just the first, simplest, portion of the logistic map ($1 < r < 3$):



In order to understand algebraically *why* this part of the logistic map looks the way it does, we must first express the recursive sequence as a function.

So, we go from this:

$$x_{n+1} = rx_n(1 - x_n)$$

To this:

$$f(x) = rx\,(1 - x\,)$$

With this new function, any input x will output the <u>next term</u> of the sequence. Moreover, since we are looking at the *long term* (very large values of n) behavior of the sequence, we know from the logistic map that for $1 < r < 3$ the sequence will settle down to ONE number. From this, we can conclude that in the long term, any term will be equal to the <u>next term</u>. In other words, any input x will be equal to its output.

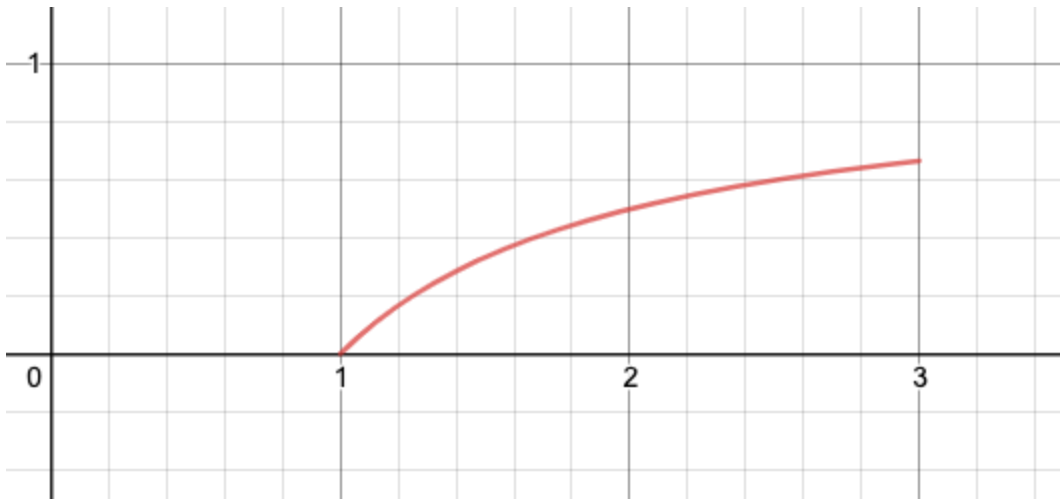
Thus, we can rewrite the function like this:

$$x = rx\,(1 - x\,)$$

If we solve for *x:*

$$x = \frac{r-1}{r}$$

If we graph this on Desmos, we see that it looks like the beginning of our logistic map!



This is because it *is*! For any *r*-value between 1 and 3 that we plug into that equation, the equation will output the value of the one term that the sequence settles to.
For example, if we plug in $r = 2.5$ to the function, we get $x = 0.6$. If we run the sequence for $r = 2.5$ on my program, we can see in the console that the sequence does in fact settle down to 0.6.

We can also algebraically see the period 2 cycle with the same method!
Previously, we said that $f(x) = x$, because we know that the sequence settles down to 1 number (any term will be equal to the next term). However, since with period 2, we know that the sequence settles down to a 2-number cycle, we can say that $f(f(x)) = x$. This equation states that any term will be equal to the term *after* the next term, which we know to be true for the period 2 cycle.

Here's the algebra for the period 2 cycle:

$$x = f(f(x))$$

$$x = r * f(x) * (1 - f(x))$$

$$x = r(rx(1 - x))(1 - (rx(1 - x)))$$

If we graph that equation on Desmos, it gives us the period 2 region of the logistic map!

We can also see that, as seen in the logistic map, the first bifurcation happens at $r = 3$.

In theory, we could repeat this process for each subsequent region of the logistic map. For example, the period 4 region could be graphed using this equation:

$$x = f(f(f(f(x)))$$

<u>Significance of the Logistic Map</u>

We saw that, for certain *r*-values, our "simple" recursive sequence produced very complicated and seemingly random behavior. The study of this type of behavior is called chaos theory. The logistic map was one of the earlier examples of how simple equations can yield chaotic behavior (in fact, it was one of the first methods of using computers to generate "random" numbers). This idea - that very small changes within a system can result in drastic differences - lies at the heart of chaos theory. We saw in the logistic map that as we changed the value of *r*, there were substantial changes in the behavior of the sequence. This idea of minor changes yielding very different results is often referred to as the Butterfly Effect -- the idea that a butterfly flapping its wings in China could cause a storm in New York the next week.

Chaos theory is an interdisciplinary field, meaning that it applies to a wide range of studies. One example of this is irregular heartbeat patterns: scientists performing a study on heart fibrillations (irregular patterns of heartbeats) in rabbits were actually able to use chaos theory to determine when to deliver electric shocks to return the rabbits' heartbeats back to normal rhythms. The irregular rhythm of water dripping from a faucet can be modeled using chaos theory. A boiling pot of water is yet another example of chaos in the real world. There's more: clouds, blood vessels, moth populations, and even the stock market can all be better understood using chaos theory. Chaos is all around us.